



# DNA Methylation Analysis: Statistics and Machine Learning in Breast Cancer

Thesis Work Presentation

Master's Degree in Mathematical Engineering

**Elisabetta Roviera**

November 27, 2025





# Table of Contents

## 1 Introduction & Motivation

### ► Introduction & Motivation

### ► Data & Computational Challenges

### ► Exploratory Data Analysis (EDA)

### ► Feature Engineering & Modeling

### ► Conclusions



# Introduction: Goals of the Presentation

## 1 Introduction & Motivation

- Background in **Mathematical Engineering** with a focus on **Data Analysis**.
- Goal: present a concrete thesis project integrating **mathematics**, **computer science**, and **biomedicine**.
- Case study: *DNA methylation* analysis in **breast cancer**.

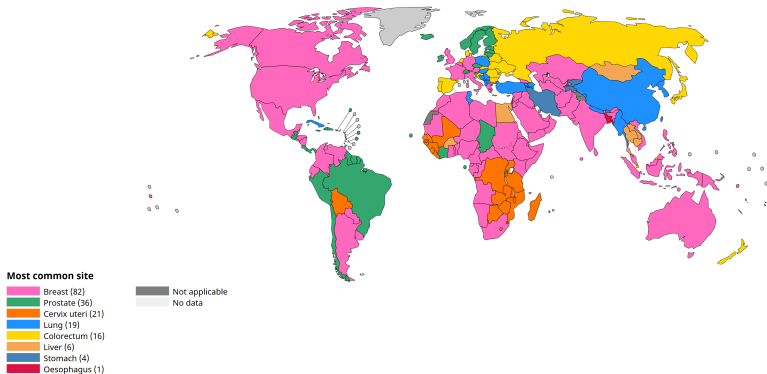
A quantitative perspective on a real biomedical dataset.



# Breast Cancer: Context and Relevance

## 1 Introduction & Motivation

- **High global incidence**, affecting millions of individuals.
- **Early detection** strongly improves patient outcomes.
- Why data matters: quantitative analysis supports **screening**, **diagnosis**, and **risk assessment**.



Most common cancer site per country, 2022 (excl. NMSC).

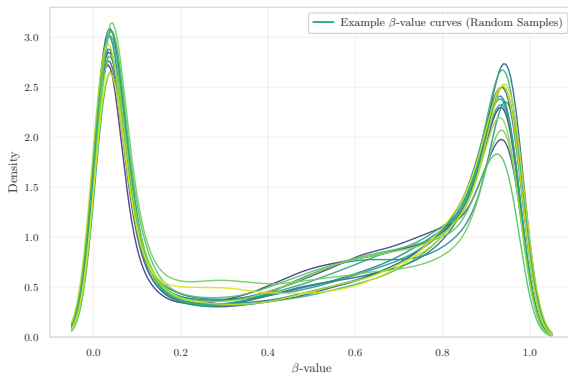
Source: Cancer TODAY — IARC, International Agency for Research on Cancer Globocan 2022.



# CpG and $\beta$ -values: The Language of Epigenetic Data

## 1 Introduction & Motivation

- **CpG sites:** genomic positions where DNA methylation can occur.
- $\beta$ -value  $\in [0, 1]$ : proportion of methylation at each CpG.
- **DNA methylation** modulates **gene regulation**.
- In cancer, both **hypermethylation** and **hypomethylation** emerge.



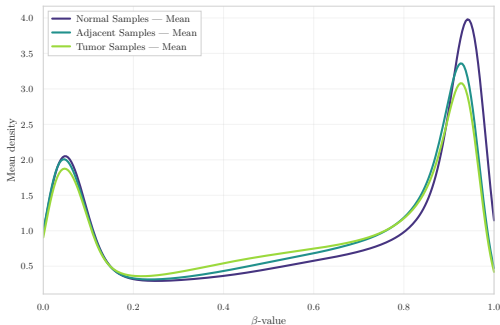
Example  $\beta$ -value density curves from random samples. Internal dataset visualisation (thesis exploratory analysis).



# Field Cancerization: The Central Research Question

## 1 Introduction & Motivation

- **Central question: is “normal” tissue near the tumor already epigenetically altered?**
- As Dr. Gambino highlights:  
*“Identify early methylation alterations in normal tissues and assess their role in tumor initiation.”*
- Concept of **field cancerization**.



Mean  $\beta$ -value densities for Normal, Adjacent, and Tumor samples. Internal dataset visualisation (exploratory analysis).

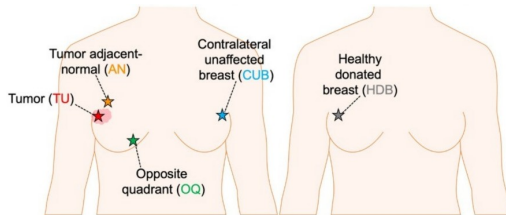


Illustration of sampling locations in breast tissue. Conceptual illustration of tumor, adjacent, and healthy breast tissue.



# Overall Research Workflow

## 1 Introduction & Motivation

- **Consultation of medical, statistical, and computational literature** to understand biological context and methodological constraints.
- **Reproduction of key results from published studies** to confirm the consistency and reliability of each dataset.
- **Progressive methodological innovation**: integrating statistical filtering, stratification strategies, dimensionality reduction, and ultimately ML/DL modelling with careful hyperparameter tuning.

**Goal:** build a robust, reproducible, and biologically-consistent analysis pipeline.



# Table of Contents

## 2 Data & Computational Challenges

► Introduction & Motivation

► Data & Computational Challenges

► Exploratory Data Analysis (EDA)

► Feature Engineering & Modeling

► Conclusions





## Accessing and Understanding the Data

### 2 Data & Computational Challenges

- GEO datasets: **heterogeneous formats** and **incomplete metadata**.
- **Small  $n$ , huge  $p$**  (up to  $\sim 800k$  CpG sites).
- Raw **IDAT files** vs. preprocessed  **$\beta$ -value matrices**.
- **Unknown or inconsistent preprocessing** in some datasets.

| Dataset   | Samples ( $n$ ) | CpGs ( $p$ ) |
|-----------|-----------------|--------------|
| GSE69914  | 397             | 485,512      |
| GSE225845 | 477             | 750,426      |
| GSE287331 | 311             | 702,573      |

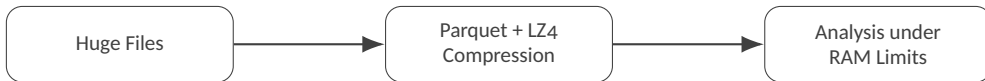
Summary of the three GEO datasets used in the analysis.



## Computational Challenges

2 Data & Computational Challenges

- **Huge files** → use of **Parquet** format, **LZ4** compression, chunked loading.
- **Strict RAM constraints** on Kaggle environments.
- Need for **Illumina manifest alignment** to correctly map probe IDs.



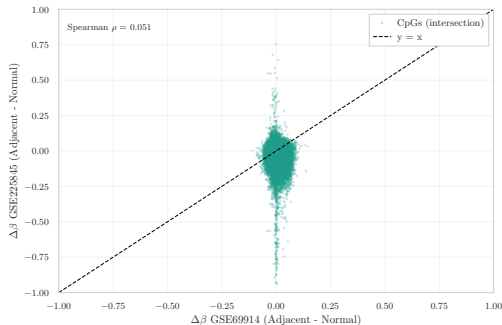
**Memory-aware processing pipeline.** From raw huge files to compressed Parquet storage, enabling analysis under strict RAM limits.



## Can I Merge the Datasets? NO

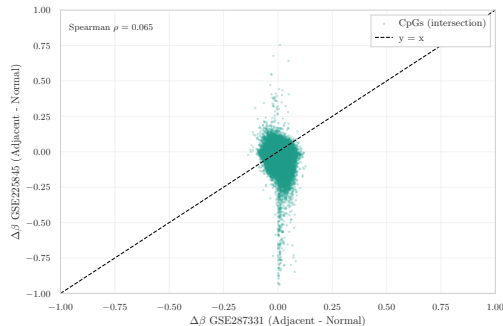
### 2 Data & Computational Challenges

- Different array platforms (HM450 vs. EPIC).
- **Heterogeneous** and partly **unknown preprocessing pipelines**.
- Direct merge would introduce **strong batch and platform effects**.
- Decision: perform all analyses **within each dataset** (intra-dataset).



$\Delta\beta$  correlation between GSE69914 and GSE225845.

Intersection CpGs show minimal concordance across platforms.



$\Delta\beta$  correlation between GSE287331 and GSE225845.

Cross-dataset signal remains inconsistent, confirming strong batch effects.



## Table of Contents

### 3 Exploratory Data Analysis (EDA)

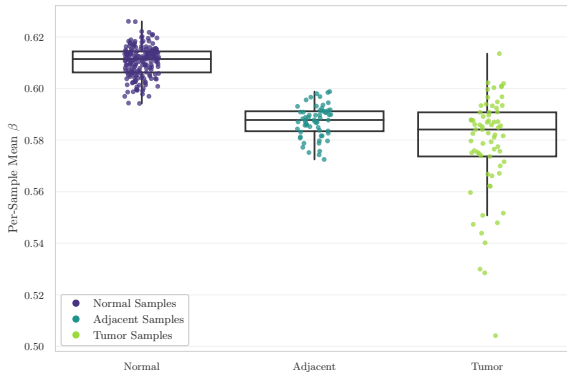
- ▶ Introduction & Motivation
- ▶ Data & Computational Challenges
- ▶ **Exploratory Data Analysis (EDA)**
- ▶ Feature Engineering & Modeling
- ▶ Conclusions



## Global DNA Methylation Distributions

### 3 Exploratory Data Analysis (EDA)

- Canonical bimodal  $\beta$ -value distribution (unmethylated vs. highly methylated CpGs).
- **Tumor samples:** shift towards **hypomethylation** and increased heterogeneity.
- **Adjacent samples:** early, subtle deviations from the Normal baseline.



Per-sample mean  $\beta$ -value distributions across tissue groups.

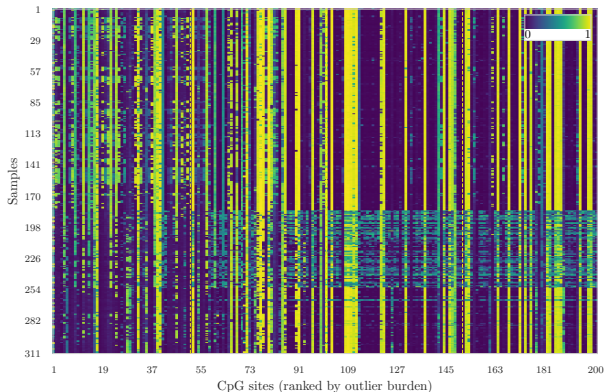
Tumor samples show global hypomethylation and higher variability compared to Normal and Adjacent samples.



## Variability and Epigenetic Outliers

### 3 Exploratory Data Analysis (EDA)

- **Unstable CpGs** are identified as those showing the highest number of extreme methylation values across samples.
- These CpGs highlight regions showing **focal hyper- or hypomethylation**.
- Local methylation instability is a hallmark of **biologically relevant epigenetic alterations**.



Heatmap of the 200 most unstable CpGs.

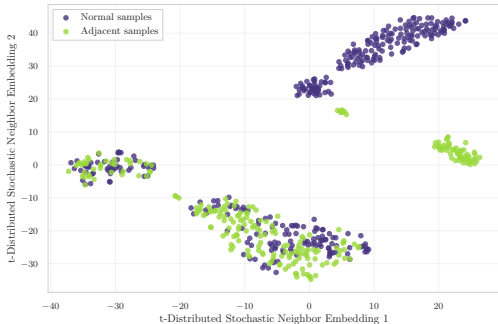
Colours represent  $\beta$ -values (blue = low methylation, yellow = high methylation).



## Dataset Embeddings and Group Structure

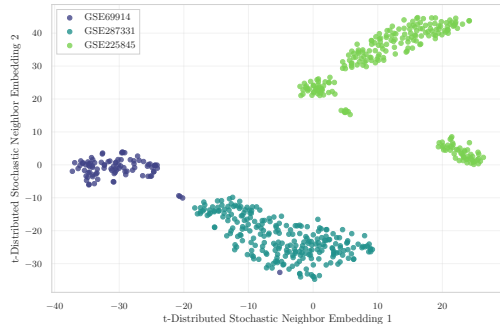
### 3 Exploratory Data Analysis (EDA)

- **GSE69914**: strong overlap between tissue groups.
- **GSE225845**: more complex and diffuse embedding structure.
- **GSE287331**: clear separation between **Normal** and **Adjacent**.



**t-SNE embedding colored by tissue state.**

Shows differences between Normal and Adjacent samples.



**t-SNE embedding colored by dataset.**

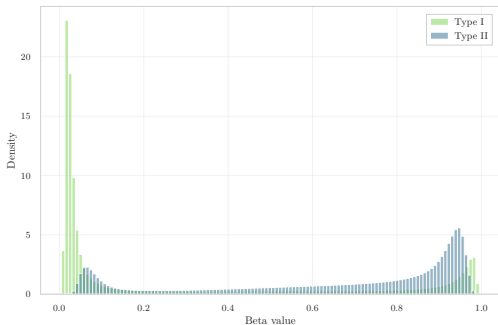
Highlights strong dataset-specific clustering and batch effects.



## Probe-Design Bias and BMIQ Normalization

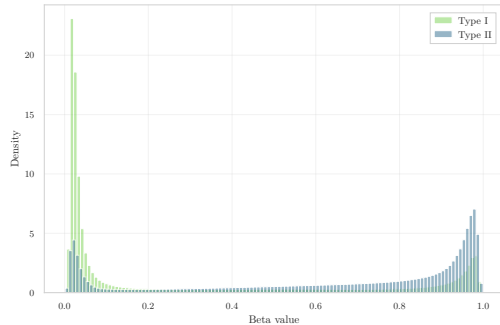
### 3 Exploratory Data Analysis (EDA)

- Two probe designs (**Type I** vs. **Type II**) with incompatible  $\beta$ -value distributions.
- **GSE287331** shows a strong uncorrected probe-design bias.
- **BMIQ** performs quantile mapping of Type II probes toward the Type I distribution.
- **Caution: BMIQ may reduce biological variability and partially mask real signals.**



**Pre-BMIQ: Type I vs. Type II distributions.**

Strong mismatch between probe types highlights probe-design bias.



**Post-BMIQ: Type I vs. Type II distributions.**

BMIQ aligns Type II probes to Type I, but may dampen biological signal.





## Table of Contents

### 4 Feature Engineering & Modeling

- ▶ Introduction & Motivation
- ▶ Data & Computational Challenges
- ▶ Exploratory Data Analysis (EDA)
- ▶ Feature Engineering & Modeling
- ▶ Conclusions



## Feature Selection in High Dimensions

### 4 Feature Engineering & Modeling

- $p \gg n \rightarrow$  **dimensionality reduction** and filtering are essential.
- Technical filters: cross-reactive probes, SNP probes, non-CpG probes, Illumina manifest.
- Evaluate and correct potential batch effects (e.g., **COMBAT**).
- Three families of feature selection strategies:
  - **Statistical**:  $\Delta\beta$ , variance filtering, stability measures.
  - **Biological**: promoters, CpG islands, curated gene sets.
  - **ML-based**: Lasso, decision trees, embedded model selection.



## ML in Non-Ideal Conditions

### 4 Feature Engineering & Modeling

- **Train / validation / test** with extremely small  $n$  (tens of samples).
- Generalising across datasets ( $A \rightarrow B$ ) is often **unreliable** due to batch and platform effects.
- **How many features to select? Which metrics are stable in the  $p \gg n$  regime?**
- **Overfitting**, noise, and limited interpretability of complex models.
- Deep Learning: training from scratch is **infeasible**  $\rightarrow$  only **pre-trained / transfer learning** approaches are realistic.

| Approach    | Data need   | Interpretability | Robustness (small $n$ ) | Typical use-case here                       |
|-------------|-------------|------------------|-------------------------|---------------------------------------------|
| Statistical | low-medium  | high             | high                    | $\Delta\beta$ , variance, QC summaries      |
| ML          | medium-high | medium           | medium-low              | classification, feature selection           |
| DL          | very high   | low              | low                     | transfer learning / pre-trained models only |

**Comparison of modelling approaches under small- $n$  / large- $p$  constraints.** Summary of data requirements, interpretability, robustness, and typical use-cases in this thesis.



# Table of Contents

5 Conclusions

- ▶ Introduction & Motivation
- ▶ Data & Computational Challenges
- ▶ Exploratory Data Analysis (EDA)
- ▶ Feature Engineering & Modeling
- ▶ **Conclusions**



## Take-home Message

### 5 Conclusions

- **Real-world bioinformatics pipeline:** from noisy GEO data and incomplete metadata to a **clean, analysable** methylation matrix via file conversion, manifest alignment, probe filtering, and batch assessment.
- **Epigenetics is deeply quantitative:** hundreds of thousands of CpGs,  $p \gg n$ , non-Gaussian  $\beta$ -distributions, strong batch and probe-design effects.
- **Role of mathematics and statistics:** exploratory analysis, stability and variability measures, dimensionality reduction, feature selection, and careful model evaluation under small- $n$ /large- $p$  constraints.
- **Machine learning in non-ideal conditions:** limited data, strong regularisation, preference for simpler and interpretable models; deep learning only via **transfer learning** or as a complement, not as a starting point.

**Big picture:** working at the interface of **biology, statistics, and computing** is challenging but extremely rewarding — and research projects like this are **much closer to our skill set than they might appear**.



# Thank you!

If you have any questions, I'd be happy to answer.